

Cell Structure & Intracellular Organelles: An Overview

Slide Companion · Current Week 2 handout controls scope · Official learning objectives guide emphasis

4 learning objectives

Organelle function

Membrane asymmetry

Transport

Na⁺/K⁺-ATPase

Learning Objectives

FACULTY EMPHASIS RULE FROM THE HANDOUT

The handout states that blue-font concepts are important, bold material is high-yield, and the narrative under the slides is intentional study content. This companion therefore treats the narrative as examinable when it supports an objective.

LO 1. Describe basic human-cell ultrastructure, identify the major organelles and cytoplasmic components, and connect each organelle to its chief function.

LO 2. Explain plasma-membrane lipid asymmetry, how it is generated and maintained, and why disruption matters.

LO 3. Classify molecules with restricted membrane diffusion and compare simple diffusion, facilitated diffusion, and active transport.

LO 4. Explain Na⁺/K⁺-ATPase function, the gradients it generates, and the cellular processes powered by those gradients.

Learning Objectives

1. Describe the basic ultrastructure of a prototypical human cell, identify the chief organelles and major components of the cytoplasm. Briefly describe the function of nucleus, mitochondria, ribosome, RER, SER, Golgi, lysosomes, peroxisomes, proteasomes, centrosome/centriole, and plasma membrane.
2. Explain membrane lipid asymmetry and explain how the asymmetry is generated and maintained. Discuss the importance of this feature for cell function.
3. Classify molecules with restricted diffusion across cell membranes and compare and contrast the mechanisms for transport of solute across the cell membrane: simple diffusion, facilitated diffusion, and active transport.
4. Explain the function of Na^+/K^+ -ATPase and its utility in generating ion gradients across cell membrane and list the biological roles of the ion gradients in cellular function.

2

Please note that I provide a detailed narrative for most of slides in all of my handouts. I will not be verbally mentioning everything that is in the narrative part verbatim in the classroom. The concepts in **blue font are important** and the ones in **bold are high-yield**. The ones in **purple is me talking to you** when I am not next to you when you are studying my handouts! Any other color is just to make sure that I grab your attention on that information!!

After you finish studying the handout please come back to the LOs page and make sure that you can answer each of the LOs.

Current handout page 2. The four objectives define the lecture. The remainder of this companion is organized around them.

Prototypical eukaryotic cell

- The cell is organized into **nucleus + cytoplasm**; the fluid part of the cytoplasm is the **cytosol**.
- Cytosol is about **pH 7.2**, mildly more acidic than extracellular fluid at about **pH 7.4**.
- In this handout, the **nucleus and nucleolus** are the organelles emphasized as visible by light microscopy; most other organelles require electron microscopy.
- Membrane-bound organelles create controlled microenvironments that can differ markedly from the cytosol.
- A relatively increased **nuclear-to-cytoplasmic ratio** is associated with malignant cells.

Chromatin

- **Nucleosome:** DNA wrapped around an octamer of histones, two each of H2A, H2B, H3, and H4.
- **Heterochromatin:** condensed and transcriptionally silent; associated with DNA methylation and histone deacetylation.
- **Euchromatin:** less condensed and transcriptionally active; histone acetylation and DNA demethylation favor accessibility.
- **Nucleolus:** rRNA transcription, pre-rRNA processing, and ribosomal-subunit assembly.
- **Nuclear pores:** regulated import/export routes for RNA, ribosomal subunits, transcription factors, and other proteins.

MEMORY HOOK

Euchromatin = accessible/expression. Heterochromatin = highly condensed. The nucleolus is the ribosome-production center.

Nucleus

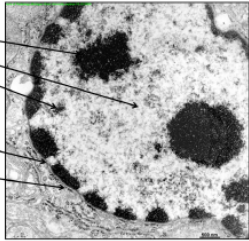


Image credit: http://hel-microscopyweb.hc.msu.edu/histology-COM/Cytology/CY_EM_Org_02_a.html

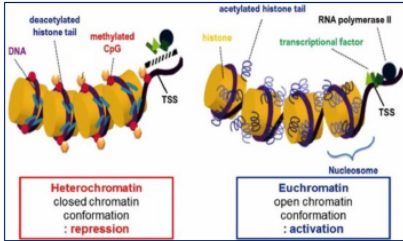
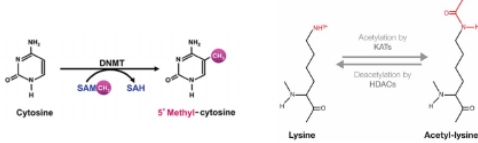


Image credit: Kim, Y.Z. Brain Tumour Res Treat. 2014; 2:7-21

Which alphabet corresponds to:

- Outer nuclear membrane
- Heterochromatin
- Euchromatin
- Nucleolus
- Nuclear pore complex



4

NUCLEUS is the largest organelle of the cell, enclosed in a double-layered nuclear membrane. While some cells, such as red blood cells, do not have a nucleus, and others, such as skeletal muscle cells, have multiple nuclei, most eukaryotic cells have a single nucleus. The genetic material of eukaryotic cells (which, in a normal somatic human cell, is 46 DNA molecules/chromosomes) is compartmentalized within the nucleus. The DNA is complexed with proteins called histones to form chromatin, the primary building block of which is called the nucleosome. Nucleosome is complex of 8 molecules of histones (2 molecules each of H2A, H2B, H3 and H4) around which DNA is wound; appear as beads on a string under electron microscope.

The chromatin is a highly dynamic structure that can be found in two interconvertible forms within the nucleus –

HETEROCHROMATIN is the highly condensed and tightly packaged form of chromatin; packing is achieved via mechanisms involving DNA methylation, histone deacetylation, and interaction with non-coding RNA; the condensed DNA is inaccessible to transcription factors and, thus, rendered transcriptionally silent.

EUCHROMATIN is less condensed, more accessible, and therefore transcriptionally active. The **condensed heterochromatin** can be **converted to the euchromatin** primarily by histone acetylation and DNA demethylation. Pharmacological drugs that inhibit **DNA methyl transferase**, the enzyme that methylates DNA are used in the **treatment of certain cancers**.

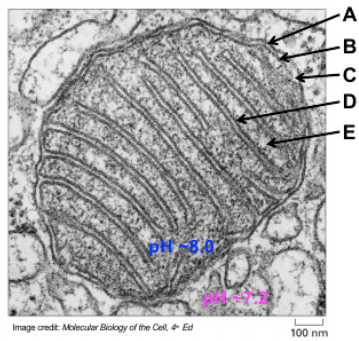
The most prominent substructure within the nucleus is the NUCLEOLUS, a small dense spherical structure, which is the **site of ribosomal RNA (rRNA) transcription, pre-rRNA processing and ribosome subunit assembly**. DNA segments that encode rRNA genes are looped into the nucleolus where it gets transcribed. The structure and the number of nucleoli in a human cell is variable and is directly related to its functions in ribosome biogenesis (image URL: <https://www.chegg.com/homework-help/questions-and-answers/describe-key-features-structure-including-function-organization-cellular-location-important-q34442531>). **Cells active in protein synthesis such as liver cells and the highly proliferative cancer cells usually tend to have a greater number of nucleoli/cell.**

Small protein-lined openings, called NUCLEAR PORES, dot the nuclear envelope. Through these pores molecules such as RNAs as well as ribosomes assembled in the nucleolus are exported out of the nucleus and transcription factors and other proteins are imported into the nucleus. The export and import of substances out and into the nucleus is highly regulated and requires many accessory proteins.

Current handout page 4. Nuclear ultrastructure, chromatin states, nucleolus, and nuclear-pore transport are all explicitly developed in the current handout.

Structure	Core function in this deck	High-yield discriminator
Mitochondria	Aerobic ATP generation; Krebs cycle, electron transport, oxidative phosphorylation; additional metabolic pathways.	Double membrane; matrix about pH 8.0; own circular DNA and ribosomes; high-energy tissues are most vulnerable to dysfunction.
Rough ER	Production and processing of membrane-embedded and secreted proteins.	Ribosome-studded surface; ER lumen is oxidizing, Ca ²⁺ -rich, and topologically equivalent to extracellular space.
Smooth ER	Lipid metabolism, drug/poison detoxification, Ca ²⁺ storage.	Sarcoplasmic reticulum is specialized SER for Ca ²⁺ storage in muscle.
Golgi	Protein processing, modification, sorting, and packaging.	Cis receives from ER; trans ships. Current deck lists glycosylation, sulfation, phosphorylation, and limited proteolysis.
Lysosome	Acid hydrolase-mediated degradation of endocytosed, membrane, and autophagic material.	Lumen about pH 4.7. Deck links targeting/enzyme defects to I-cell disease and Gaucher disease.
Peroxisome	Very-long-chain fatty-acid beta-oxidation, branched-chain fatty-acid alpha-oxidation, free-radical handling.	Catalase converts H ₂ O ₂ to H ₂ O. Deck links defects to X-linked adrenoleukodystrophy and Zellweger syndrome.
Ribosome	Protein synthesis.	Prokaryotic 70S = 30S + 50S; eukaryotic 80S = 40S + 60S.
Proteasome	Quality control and degradation of endogenous cytosolic proteins.	Polyubiquitination marks proteins; regulatory particles use ATP to unfold substrate; bortezomib inhibits proteasome function.
Centrosome/centriole	Microtubule organization and spindle formation; centrioles can become basal bodies.	Centriole = nine triplet microtubules. Primary cilium defects can cause ciliopathies such as polycystic kidney disease.

Mitochondria



Which alphabet corresponds to: the matrix, a crista, the intermembrane space, the outer membrane, and the inner membrane?

Where would you find the DNA & the ribosomes?

- Inner mitochondrial membrane is rich in cardiolipin
- The number of mitochondria in a cell varies significantly

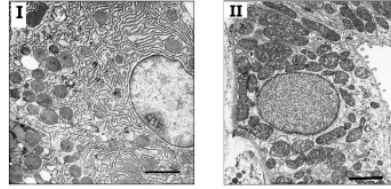


Image credit: https://ihl-microscopyweb.hc.msu.edu/Histology-COM/Cytology/CY_EM_Ob_03_02_a.html

Which of these two images is from a cell that is involved in reabsorption of nutrient by an energy-requiring active transport process?

5

MITOCHONDRIA are rod-shaped double membrane-bound organelles that are distributed throughout the cytosol of most eukaryotic cells – except RBCs. Main function is the conversion of the potential energy of food molecules into ATP under aerobic conditions through a set of metabolic pathways – the Krebs cycle, electron transport chain and oxidative phosphorylation. Their number within the cell ranges from a few 100 to 1000s in very active cells such as the muscle cells, nerve cells, and kidney cells. Thus, a patient with **defective mitochondrial function** will display **disease symptoms in the muscle and the brain** and, sometimes, in the kidneys as well.

Mitochondria have a highly 'porous' outer membrane and an inner membrane that is highly impermeable to many ions, including protons, and convoluted so that a large number of infoldings called cristae are formed. The space within the inner membrane is called the **matrix (pH ~8.0, which is significantly alkaline compared to the ~7.2 pH of the cytosol)**. The mitochondria is also the 'home' to many important metabolic pathways such as fatty acid oxidation, urea synthesis, heme synthesis, steroid metabolism, etc.

Compared to other cellular organelles, the mitochondrion is unique because it contains **5 – 10 circular molecules of DNA** and several ribosomes in the matrix. Each mitochondrial DNA molecule consists of ~16.5 kilo base pairs carrying the information for 37 genes which encode: 2 different molecules of ribosomal RNA (rRNA), 22 different molecules of transfer RNA (tRNA) (at least one for each amino acid) and 13 proteins. The rRNA and tRNA molecules are used to synthesize the 13 proteins that **participate in the formation of 4 of the 5 complexes (complexes I, III, IV, & V)** of the electron transport chain. Each of these protein complexes also requires subunits that are encoded by nuclear genes, synthesized on free ribosomes (not membrane-bound), and imported from the cytosol into the mitochondrion. There are an additional ~1,000 other proteins encoded by the nuclear genes that must be imported into the mitochondrion that participate in the other metabolic functions mentioned earlier. **Thus, mutations in both nuclear DNA as well as mitochondrial DNA can give rise to heritable genetic diseases, however, these diseases show different inheritance patterns.**

Mitochondrial inheritance

- Mitochondrial DNA is transmitted through the **maternal line** in the deck.
- An affected male does not transmit a mitochondrial-DNA disorder to offspring.
- **Heteroplasmy** explains variable mutation load and variable severity between tissues and among children.
- Nuclear-gene defects that impair mitochondrial proteins can produce similar organ dysfunction but do **not** follow maternal mitochondrial inheritance.

Lysosome & Peroxisome

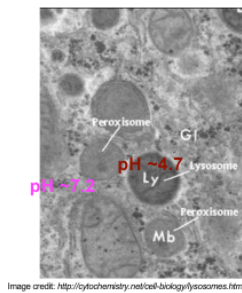


Image credit: <http://lysochemistry.net/cell-biology/lysosomes.htm>

Clinical Relevance

Improper targeting of enzymes to the lysosomes or genetic defects in lysosomal enzymes can cause serious lysosomal diseases such as I-cell or Gaucher's

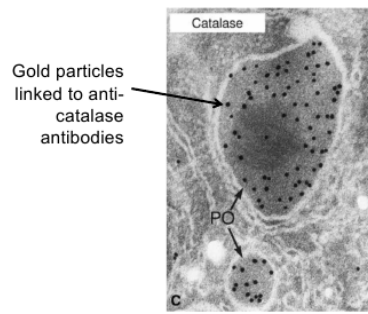


Image credit: Histochem Cell Biol 2008 Apr; 129(4): 421-440.

Clinical Relevance

Genetic defects in peroxisomal enzymes is the cause of X-linked adrenoleukodystrophy & defect in import of enzymes cause Zellweger syndrome

9

Lysosomes & peroxisomes are similar looking membrane organelles. However, they have different origin and function. While **lysosomes originate from Golgi**, **peroxisomes do not** and is not well understood. Peroxisomes are distinguished by a crystalline structure inside a sac which also contains amorphous gray material.

LYSOSOMES are filled with hydrolytic enzymes are responsible for degrading molecules within the cell. These membrane-bound organelles contain a variety of enzymes called hydrolases that can digest proteins, nucleic acids, lipids, and complex sugars. These **enzymes are synthesized in RER** and get processed and **packaged in Golgi**. The **luminal content of a lysosome is more acidic (~pH 4.7) than the cytoplasm (~pH 7.2)**. This environment activates the hydrolases and confines their destructive work to the lysosome. They can fuse with endosomes, creating so-called endolysosomes (or phagolysosomes), and degrade the material within the endosome. Lysosomes primarily hydrolyze extracellular proteins (e.g., plasma proteins), that are taken into the cell by endocytosis, cell-surface membrane proteins that are used in receptor-mediated endocytosis, as well as the proteins (and other macromolecules) engulfed by autophagosomes. Lysosomes also help to defend against pathogen entry via endocytosis by degrading pathogens before they reach the cytoplasm via an oxygen-independent killing mechanism.

PEROXISOMES are somewhat similar to mitochondria in function. They are small organelles that contain enzymes that are linked to various metabolic pathways such as **β -oxidation of very-long chain-fatty acids and α -oxidation of branched-chain fatty acids**. Like the nuclear-encoded mitochondrial enzymes, the **peroxisomal enzymes are synthesized on free ribosomes in the cytosol**, following which they are imported into the lumen of peroxisomes. The size and enzyme composition of peroxisomes vary considerably, but all contain enzymes that use molecular oxygen to neutralize free radicals, **producing large amounts of toxic hydrogen peroxide (H_2O_2)** in the process. Catalase, a peroxisome-localized enzyme, efficiently decomposes H_2O_2 into H_2O . Peroxisomes are most abundant in liver cells, where they constitute about one to two percent of the cell volume.

Current handout page 9. Lysosome and peroxisome are intentionally contrasted in the handout by pH, enzymes, metabolic role, and associated disease examples.

DO NOT MISS

The current handout explicitly states that **circulating fully differentiated red blood cells retain proteasomes but no other organelles**. That is a favorite one-line discriminator because it combines cell structure with RBC physiology.

Leaflet composition

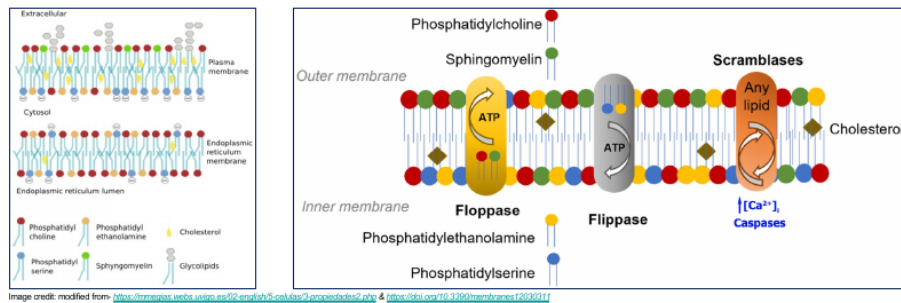
- **Inner/cytosolic leaflet:** higher phosphatidylserine (PS), phosphatidylethanolamine (PE), and phosphatidylinositol (PI).
- **Outer leaflet:** higher phosphatidylcholine (PC), sphingomyelin (SM), and glycolipids.
- Carbohydrates are also predominantly displayed on the outer surface.

Translocase	Direction/function in current deck	ATP?
Flippase	Moves selected phospholipids toward the inner/cytosolic monolayer and helps create/maintain asymmetry.	Energy-dependent
Floppase	Moves polar lipids toward the outer/extracellular or luminal monolayer; deck identifies these as ABC-family transporters.	Energy-dependent
Scramblase	Bidirectional lipid exchange that dissipates asymmetry.	No ATP in the deck

WHY ASYMMETRY MATTERS

PS exposure on the outer surface can mark apoptotic cells for phagocytic clearance. In platelets/erythrocytes, calcium-dependent scrambling can also generate a **procoagulant surface**.

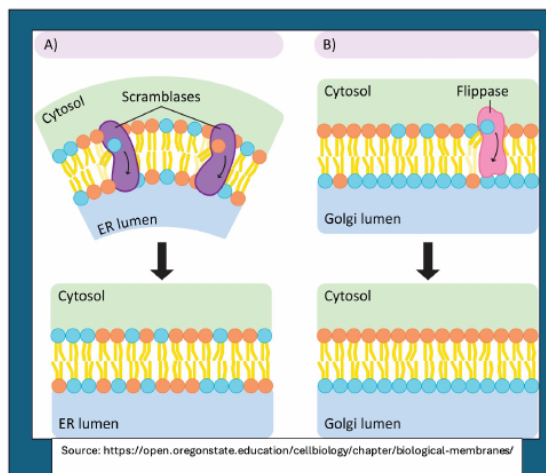
LO 2: Membrane Asymmetry



- PS, PE, PI are higher in the inner leaflet; PC, SM, GL are higher in the outer leaflet
- Exposure of PS at the cell surface is triggered by physiological prompts that activate plasma membrane scramblases
- Surface display of PS provides several functions depending on the physiological context, including marking apoptotic cells for consumption by phagocytes

14

In plasma and organelle membranes there is one monolayer facing the extracellular space or the lumen of the organelle, respectively, whereas the other monolayer usually faces the cytosol. Though carbohydrates (which are mostly present on the outer membrane) and proteins are differentially distributed between these two monolayers of the membrane, the term membrane asymmetry is largely related to the lipid distribution, especially the phospholipids, across the lipid bilayer in the plasma membrane (usually not seen in other membranes such as the ER membrane).



The majority of the **negatively-charged lipids** such as **phosphatidylethanolamine**, **phosphatidylinositol** and **phosphatidylserine** are located in the **inner leaflet** whereas most of the **positively-charged lipids** having **choline**, such as **phosphatidylcholine** and **sphingomyelin** are **localized predominantly in the outer leaflet**. Disorganization of the membrane asymmetry usually signals damage to the cell. For example, during apoptosis (programmed cell death), the outer monolayer of the plasma membrane exhibits lipids that are

Current handout page 15. The current deck directly maps inner versus outer leaflet lipids and links phosphatidylserine externalization to scramblase activation and apoptosis.

What crosses a naked lipid bilayer?

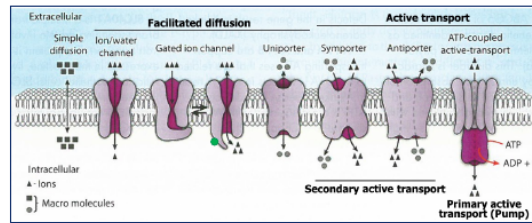
Gases and lipophilic molecules cross most readily. Hydrophilic nutrients, waste products, and ions require membrane proteins to cross efficiently.

Mechanism	Protein?	Energy?	Direction	Key behavior
Simple diffusion	No transport protein required	No	Down concentration gradient	Not carrier-limited
Facilitated diffusion	Channel or carrier	No	Down concentration gradient	Specific; carrier-mediated transport can saturate and can show competitive inhibition
Primary active transport	Pump/transporter	Direct ATP	Can move solute against gradient	Na ⁺ /K ⁺ -ATPase is the central example
Secondary active transport	Cotransporter	Uses stored electrochemical-gradient energy	One solute can move against its gradient	Symport = same direction; antiport = opposite directions

Channel versus carrier

- **Channels** form pores and allow very rapid ion movement down an electrochemical gradient. The deck distinguishes voltage-gated, ligand-gated, and mechanically gated channels.
- **Carriers/transporters** bind a solute and change conformation. Their finite number produces **saturation**.
- The deck uses CFTR as an ATP-gated anion-channel example and a sodium-dependent glucose cotransporter as a secondary-active symporter example.

Integral Membrane Proteins Make the Membrane Selectively Permeable



Conformational shape change of the transporter before and after interaction with the solute molecule or a ligand transports solute from one side of the membrane to the other

Image credit: modified from Integrative Medical Biochemistry, King, McGraw Hill

16

Specific **integral membrane proteins** make the plasma membrane to be **selectively permeable** and allow some of the solutes to get across the membrane. The type of transport mediated by these proteins can be grouped under facilitated diffusion and active transport. You can watch a very informative animation on all of the various membrane transport processes here: <https://www.youtube.com/watch?v=oxX2fq2DBBo>.

Classification of transport across cell membranes

		Require transport protein?	
		Yes	No
Requires energy?	Yes	Active transport	
	No	Facilitated diffusion	Simple diffusion

The term **FACILITATED DIFFUSION** (also called facilitated transport) is used when molecules **diffuse bidirectionally** across the plasma membrane **from an area of higher concentration to an area of lower concentration** with assistance from membrane proteins without expending cellular energy. In addition, since they are mediated by proteins, the transport process is highly specific to the solutes. E.g., most of the sugar transporters are specific to the D-isomer such as D-glucose while the amino acid transporters are specific to the L-isomer such as the L-amino acids. Since the rates of facilitated diffusion are limited by the number of functional membrane proteins, transport through facilitated diffusion can be saturated at higher concentrations of the solute. The transport proteins shield the solute molecules from the hydrophobic repulsive force of the membrane, providing a route by which they can cross. Some of these integral proteins form a pore or channel through the phospholipid bilayer and are called channel proteins. Others are transporter (or carrier) proteins which bind to the substance and aid its movement through the membrane. **Facilitated diffusion/transport unlike passive diffusion will show speed and specificity, saturation kinetics, susceptibility to competitive inhibition, and susceptibility to chemical inactivation.**

The term **ACTIVE TRANSPORT** is used when the movement of a substance across the membrane with the assistance from membrane proteins require the use of cell's energy. A characteristic feature of most of the

Current handout page 18. The handout separates facilitated diffusion from active transport and emphasizes specificity, saturation, competitive inhibition, and energy use.

PUMP STOICHIOMETRY

1 ATP -> 3 Na⁺ out + 2 K⁺ in. The pump is electrogenic because more positive charge leaves than enters.

Three gradients generated

1. **Outside-to-inside Na⁺ gradient**
2. **Inside-to-outside K⁺ gradient**
3. **Inside-negative electrical potential**

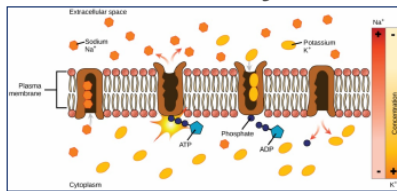
In polarized cells, the deck places the pump on the **basolateral membrane**.

Primary versus secondary active transport

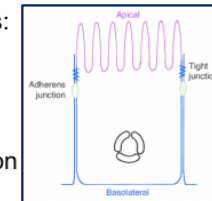
Primary active transport spends ATP directly. Secondary active transport uses energy already stored in an ion gradient. A **symporter** carries coupled solutes in the same direction; an **antiporter** carries them in opposite directions.

Primary & Secondary Active Transport

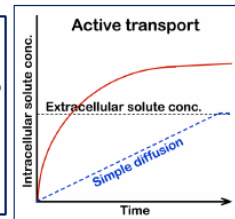
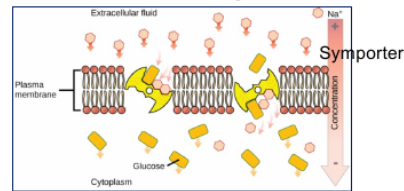
Primary



- The Na^+/K^+ -pump generates 3 gradients:
 1. Outside $\rightarrow$ inside Na^+ gradient
 2. Inside $\rightarrow$ outside K^+ gradient
 3. Inside-negative membrane potential
- In a polarized cell, the pump is located on the basolateral side of the cell



Secondary



19

PRIMARY ACTIVE TRANSPORT is **directly dependent on ATP energy** to drive the transport process and frequently called pumps. The best and most important example of an active transporter is the sodium-potassium pump (or Na^+/K^+ -ATPase). Using ATP as the energy, this transporter transports 3 Na^+ out of the cell while simultaneously transporting 2 K^+ into the cell. Based on stoichiometry, thus, for every 3 ions of sodium that move out, only 2 ions of potassium move in, resulting in a more negative cell interior. Thus, the **Na^+/K^+ -pump generates 3 distinct gradients** namely, the outside-to-inside Na^+ gradient, inside-to-outside K^+ gradient and the inside electronegative potential gradient. Watch an animation of Na^+/K^+ ATPase here: https://www.youtube.com/watch?v=_bPFKdWICg.

The process consists of the following six steps.

1. With the enzyme oriented towards the interior of the cell, the carrier has a high affinity for sodium ions. Three ions bind to the protein.
2. ATP is hydrolyzed by the protein carrier and a low-energy phosphate group attaches to it.
3. As a result, the carrier changes shape and re-orientates itself towards the exterior of the membrane. The protein's affinity for sodium decreases and the three sodium ions leave the carrier.
4. The shape change increases the carrier's affinity for potassium ions, and two such ions attach to the protein. Subsequently, the low-energy phosphate group detaches from the carrier.
5. With the phosphate group removed and potassium ions attached, the carrier protein repositions itself towards the interior of the cell.
6. The carrier protein, in its new configuration, has a decreased affinity for potassium, and the two ions are released into the cytoplasm. The protein now has a higher affinity for sodium ions, and the process starts again.

SECONDARY ACTIVE TRANSPORT (cotransport) processes, **use electrochemical gradients generated by active transporters** such as the Na^+/K^+ -pump as an energy source to move molecules against their gradient, and thus does not directly require a chemical source of energy such as ATP. There are 2 types of secondary active cotransporters called SYMPORTERS or ANTIPORTERS. When a protein transports both of the transported particles in the same direction it is called a symporter and when a transporter transports the particles in opposite direction it is called an antiporter. E.g., a secondary active transporter called sodium-

Current handout page 22. Current handout page 22 shows the six-step Na^+/K^+ -ATPase cycle, its 3:2 stoichiometry, the three generated gradients, and secondary-active cotransport.

Cell communication

Na⁺ and K⁺ gradients permit neuronal depolarization and electrical signaling.

Muscle function

Ion gradients support skeletal and cardiac muscle excitability and contraction.

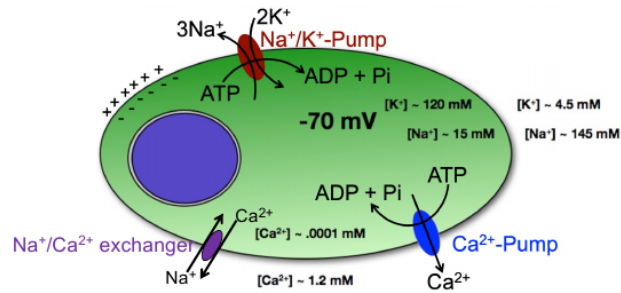
Enzyme regulation

The Na⁺/Ca²⁺ exchanger helps keep cytosolic Ca²⁺ low until signaling raises it.

Hormone secretion

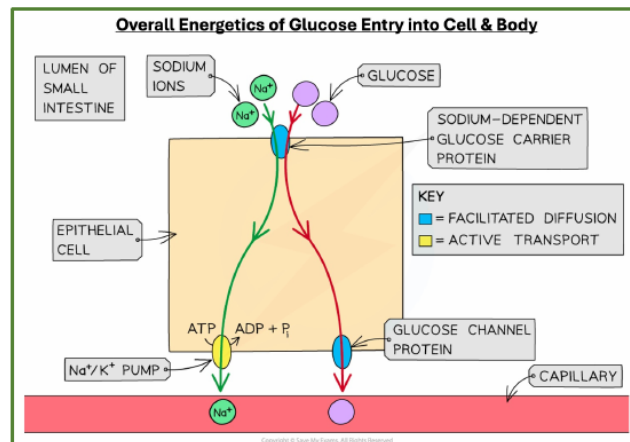
The deck explicitly links ionic gradients and Ca²⁺ signaling to secretion, including insulin release from pancreatic beta cells.

LO 4: Ion Gradients in Cell and Their Biological Function



- Cell communication: neuronal function, skeletal and cardiac muscle function
- Regulation of enzyme activity
- Secretion of hormones

20



Other Roles for the Electrochemical Gradients – Apart from driving the secondary active transport, energy from electrochemical gradients are used for many other cellular processes. One important role is in cell communication, in particular, the **communication between neurons and their targets**. Active neurons send signals down their axons by allowing sodium ions to enter the cytosol, thereby depolarizing the axon. Another important role is to **regulate the activity of enzymes** by regulating intracellular Ca^{2+} levels. The Na^+ gradient is utilized by the $\text{Na}^+/\text{Ca}^{2+}$ exchanger, an antiporter that exchanges intracellular Ca^{2+} for extracellular Na^+ thus lowering the intracellular Ca^{2+} by about 1000-fold compared to outside. Many enzymes require calcium for their activity. Because the cytosolic concentration of calcium is low, most of these enzymes are inactive. When cells want to activate these enzymes, they allow the cytosolic concentration of calcium to rise by opening a Ca^{2+} channel. Electrochemical gradients are also used to **regulate cellular**

Current handout page 24. The deck summarizes biological uses of electrochemical gradients: communication, muscle function, enzyme regulation, and hormone secretion.

Discriminations worth testing

- **Euchromatin vs heterochromatin:** accessibility, transcription, acetylation/methylation direction.
- **RER vs SER vs Golgi:** protein production, lipid/detox/ Ca^{2+} , modification/sorting.
- **Lysosome vs peroxisome vs proteasome:** acid hydrolases, VLCFA/ H_2O_2 metabolism, ubiquitin-tagged endogenous protein degradation.
- **Mitochondrial DNA vs nuclear-encoded mitochondrial disease:** maternal inheritance only applies to mtDNA defects.
- **Flippase/floppase vs scramblase:** maintained asymmetry versus bidirectional scrambling.
- **Channel vs carrier:** pore/speed/gating versus binding/conformational change/saturation.
- **Facilitated diffusion vs active transport:** both can require proteins, but only active transport uses energy to drive uphill movement.
- **Primary vs secondary active transport:** direct ATP versus gradient energy.
- **Symport vs antiport:** same direction versus opposite directions.
- **Na^+/K^+ -ATPase consequence:** 3 Na^+ out, 2 K^+ in $\rightarrow$ Na^+ gradient, K^+ gradient, inside-negative potential $\rightarrow$ downstream cellular work.

SCOPE GUARD

Details that are not supported by the current Week 2 course materials are intentionally **not promoted into this companion's core scope**. The current handout and official learning objectives remain the controlling references.

Summary

- Under light microscopy only nucleus and nucleolus are visible inside the cell; EM helps visualize lot many organelles
- Nucleolus is the ribosome 'factory' of the cell and its number may be higher in the nucleus of cancer and highly proliferating cells
- Chromatin is dynamic structure and DNA methylation and histone acetylation aid in changing chromatin structure to heterochromatin and euchromatin state, respectively
- Mitochondria are the 'powerhouse' of the cell with their own genome which resembles prokaryotic genome. Mitochondria is inherited only from the mother and mitochondrial disorders are transmitted to every child of the mother and show varying disease severity (heteroplasmy)
- Peroxisomes and mitochondria have similar function and lysosomes and peroxisomes appear similar under the EM
- Proteasomes are the intracellular protein degrading/clearing factories and are the only organelles found in mature RBCs
- The two leaflets of the plasma membrane are asymmetric with +vely charged and glycosylated phospholipids concentrated on the outer leaflet and -vely charged phospholipids such as phosphatidylserine and phosphatidylethanolamine concentrated on the inner leaflet

21

- Externalization of phosphatidylserine by Ca^{2+} -dependent scramblases triggers apoptosis
- Lipid bilayer by itself is impermeable to most solute molecules and presence of integral membrane proteins such as channels, transporters, and pumps help in cells uptake/secrete molecules
- Channels form 'pores' in the membrane and facilitate the diffusion of molecules across the membrane and help attain equilibrium across the membrane of a given solute molecule within a fraction of a second. They can be always open or voltage-, ligand-, or mechanically gated and the solute always moves from higher concentration to lower concentration
- Facilitated transporters are similar to channels facilitating movement of solutes down the concentration gradient, however, instead of allowing solutes do not form a channel and the diffusion takes place much slower than a channel.
- Active transporters which are of two types, primary and secondary, can move solute molecules across against a concentration gradient by a process which requires energy. Primary active transporters use directly ATP as the energy and are frequently called pumps.
- The best known and most important primary active transporter is the Na^+/K^+ pump which pumps 3 Na^+ ions out of the cell for 2 K^+ ions moving into the cell and consuming one ATP during this process
- Because of this stoichiometry, it generates outside to inside Na^+ gradient, inside to outside K^+ gradient, and inside negative membrane potential – three different energy gradients
- Secondary active transporters use the gradient energy developed by the Na^+/K^+ -ATPase to move solute molecules across the membrane against a concentration gradient
- In symport, both the solute moving from a higher concentration to the lower concentration and the solute being moved from the lower concentration to the higher concentration are moving in the same direction –either from inside the cell to the outside or from the outside the cell to inside. In antiport the two solutes are moving in the opposite direction
- The ionic gradients as well as the inside-negative membrane potential generated by the Na^+/K^+ -Pump will be used for various cellular function such as cellular contraction, hormone secretion, cellular communication, etc.

Current handout page 26. The handout summary is a useful final checklist because it returns directly to the organelle, asymmetry, transport, and Na^+/K^+ -ATPase objectives.

One-pass recall checklist

Can you explain?

N:C ratio, nucleosome, heterochromatin/euchromatin, nucleolus, mitochondrial maternal inheritance, heteroplasmy.

Can you contrast?

RER/SER/Golgi, lysosome/peroxisome/proteasome, channel/carrier, facilitated/active transport.

Can you draw?

Inner vs outer leaflet lipids and the direction/function of flippase, floppase, and scramblase.

Can you reason downstream?

What happens after Na^+/K^+ -ATPase creates Na^+ , K^+ , and electrical gradients?